

Complete Chapters 1 to 3 Writing Guide & Technical Content Map

Integrated Library Management System (CDM ILMS) — Zero-Mistake Manuscript Framework

Project Title: Colegio de Montalban - Integrated Library Management System (CDM ILMS)

Scope: Chapters 1, 2, & 3 Complete Content Blueprint

Target System: Desktop Librarian SPA + Mobile PWA Student Portal + Express REST API

Institutes: ICS, ITE, IBE Academic Programs

Evaluation Standard: ISO/IEC 25010 Software Quality Assurance Framework

Catalog Standard: MARC 21 Tagging + LC Call Numbers

CHAPTER 1: THE PROBLEM AND ITS BACKGROUND

1.1 Background of the Study

What to Write & Input in Manuscript:

"Academic libraries in higher education institutions serve as the vital hub for knowledge dissemination and curriculum support. At Colegio de Montalban (CDM), supporting academic programs across the Institute of Computer Studies (ICS), Institute of Teacher Education (ITE), and Institute of Business and Entrepreneurship (IBE) requires an agile library management system. Traditional manual logbook transactions and unintegrated catalog systems create operational bottlenecks, slow inventory audits, and restrict student access to physical references outside operational hours. To resolve these challenges, the Integrated Library Management System (CDM ILMS) was developed as a dual-platform web and mobile Progressive Web Application (PWA)..."

1.2 Statement of the Problem

General Problem: How to design, develop, and implement an Integrated Library Management System for Colegio de Montalban that automates book cataloging, borrowing circulation, mobile student self-service, and account security?

Specific Research Questions:

- **SOP 1:** How to implement standardized academic book cataloging compliant with MARC 21 tagging standards across ICS, ITE, and IBE institutes?
- **SOP 2:** How to automate real-time book borrowing and transaction logging with accurate copy inventory control?
- **SOP 3:** How to develop a responsive mobile Progressive Web App (PWA) allowing students to search references, make mobile reservations, and generate digital student passes?
- **SOP 4:** How to secure user accounts and transactions using JWT authentication, bcrypt password hashing, 6-digit Email OTP recovery, and 2D QR Code receipt verification?
- **SOP 5:** What is the evaluation level of the system in terms of ISO/IEC 25010 Software Quality standards (Functional Suitability, Usability, Security, Performance Efficiency)?

1.3 Objectives of the Study

General Objective: To design, build, and deploy an Integrated Library Management System for Colegio de Montalban that automates library administrative management and student mobile access.

Specific Objectives: Directly map these to your SOPs above (e.g., Objective 1 maps to SOP 1: To develop a MARC 21 cataloging engine; Objective 2 maps to SOP 2: To create a real-time borrowing ledger module; Objective 3 maps to SOP 3: To build a Mobile PWA Student Portal).

1.4 Significance of the Study

- **To Colegio de Montalban Librarians:** Streamlines inventory audits, automates overdue calculations, and provides real-time transaction reports.
- **To CDM Students:** Enables 24/7 mobile catalog access, mobile borrow reservations, and contactless QR turnstile verification.
- **To College Administration:** Provides accurate data analytics on book utilization across ICS, ITE, and IBE institutes.

1.5 Scope and Limitations

- **Scope:** Covers MARC 21 book cataloging, real-time search/filtering (useMemo, useEffect), student borrowing/returns, mobile PWA portal, 2D QR receipts, 6-digit Email OTP recovery, and role-based access control.
- **Limitations:** System requires initial network connectivity for online API data synchronization; RFID physical hardware scanners can be integrated via barcode API extensions.

CHAPTER 2: REVIEW OF RELATED LITERATURE AND STUDIES

2.1 Foreign & Local Literature Matrix

Category	Literature / Study Topic	Relevance to CDM ILMS Implementation
Foreign Literature	MARC 21 Machine-Readable Cataloging Standards (Library of Congress)	Establishes standard metadata tags (001, 020, 082, 100, 245) used in books cataloging.
Foreign Literature	Progressive Web Application (PWA) Architecture in Higher Education	Supports offline caching, service workers, and mobile web app installation without app stores.
Local Literature	Philippine Academic Library Digitalization & CHED Automation Directives	Justifies transition from manual paper ledgers to centralized web-based library automation.
Local Studies	Automated Library Circulation & QR Code Identity Verification in PH Universities	Validates 2D QR Code receipt generation for turnstile exit verification.

2.2 Conceptual Framework: Input-Process-Output (IPO) Model

Use this exact IPO structure in your manuscript diagram and text:

INPUT (Requirements & Resources)	PROCESS (System Development)	OUTPUT (Final System)
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• CDM Library Requirements • Student & Librarian Data • MARC 21 Tagging Rules • React 19, TypeScript, Tailwind • Node.js, Express.js REST API • SQLite3 & `better-sqlite3` • ISO/IEC 25010 Quality Metrics	• Agile SDLC Methodology • Database Schema Design • REST API Route Controller Coding • React Component UI/UX Build • JWT & Bcrypt Auth Integration • PWA Service Worker Setup • System Testing & ISO Audit	CDM Integrated Library Management System (CDM ILMS) • Desktop Librarian SPA • Student Mobile PWA Portal • Automated Borrowing Ledger • Email OTP Recovery Engine • 2D QR Receipt Generator
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CHAPTER 3: RESEARCH METHODOLOGY

3.1 SDLC Methodology: Agile Development Model

The project adopted an **Agile Software Development Life Cycle (SDLC)** methodology featuring iterative design, prototyping, and rapid testing. Agile was selected because it allowed continuous feedback integration from Colegio de Montalban library staff during system building.

3.2 Technical Architecture & Software Specifications

Include this exact table of technical specifications in Chapter 3:

Component Layer	Software / Tool Specification	Technical Justification & Purpose
Client Desktop SPA	React 19, TypeScript, Vite 6, Tailwind v4	Fast, type-safe single-page front-end with 60 FPS search rendering.
Client Mobile PWA	PWA Service Worker, Manifest JSON, LocalStorage	Cross-platform mobile access for students without app store installation.
Server REST API	Node.js, Express.js v5, CORS	Handles asynchronous HTTP API requests, controllers, and validation.
Database Layer	SQLite 3, `better-sqlite3` driver	Synchronous, high-speed relational database engine for transaction logs.
Authentication Engine	JSON Web Tokens (JWT), `bcrypt`	Stateless session authorization and 10-round salted password hashing.
QR Verification	`qrcode` Canvas Engine	Generates verified 2D QR Code receipts for turnstile verification.

3.3 Database Relational Schema Details

Document these exact 6 relational tables in Chapter 3 under Data Design:

1. **`books` Table:** `id` (PK), `title`, `author`, `isbn`, `category`, `cover`, `available`, `total`, `borrowCount`, `publishYear`, `callNo`, `institute`, `yearLevel`, `semester`, `marcTags`
2. **`students` Table:** `studentId` (PK), `name`, `email`, `institute`, `course`, `yearLevel`, `status`
3. **`librarians` Table:** `id` (PK), `username`, `password` (hashed), `name`, `role`
4. **`transactions` Table:** `id` (PK), `bookId` (FK), `studentId` (FK), `issueDate`, `dueDate`, `returnDate`, `status`
5. **`reservations` Table:** `id` (PK), `bookId` (FK), `studentId` (FK), `reservationDate`, `status`
6. **`otp\_codes` Table:** `id` (PK), `email`, `code`, `expires\_at`, `used`

3.4 ISO/IEC 25010 Software Evaluation Framework

Specify that the system evaluation uses Likert scale survey instruments evaluated across 5 ISO/IEC 25010 characteristics:

- **Functional Suitability:** Evaluates cataloging, borrowing, returning, and search completeness.
- **Usability:** Evaluates UI responsiveness, layout clarity, `■K` keyboard shortcut, and mobile PWA navigation.
- **Security:** Evaluates JWT authentication, `bcrypt` password hashing, and Email OTP recovery integrity.
- **Performance Efficiency:** Evaluates REST API response times (< 15ms) and `useMemo` search rendering.
- **Reliability:** Evaluates database query stability and error fallback handling in UI components.